

DSCI550: Data Science at Scale
Homework 4, Spring 2026
(Due on: 11:59 pm, April 2, 2026)

1. (20 pts) *k*-Means Clustering: For the following six points,

	X	Y
A1	1.00	2.00
A2	1.00	4.00
A3	3.00	1.00
A4	3.00	5.00
A5	5.00	2.00
A6	5.00	4.00

- 1) Use the *k*-means algorithm to show the final clustering result assuming initially we assign A1, A6 as the center of each cluster, respectively.
- 2) Use the *k*-means algorithm to show the final clustering result assuming initially we assign A3, A4 as the center of each cluster, respectively.
- 3) Compute the quality of the *K*-Means clustering using the Sum of Squared Error (SSE) which shows cohesion measures how near the data points in a cluster are to the cluster centroid. Given a set of observations $(x_1, x_2, \dots, x_n)$, where each observation is a d -dimensional real vector, *k*-means clustering aims to partition the n observations into k ($\leq n$) sets $S = \{S_1, S_2, \dots, S_k\}$ so as to minimize the intra-cluster sum of squares.

$$SSE = \sum_{i=1}^k \sum_{x \in S_i} (x - \mu_i)^2$$

where μ_i is the mean of points in S_i .

Based on SSE of 1) and 2), which clustering would be better?

2. (40 pts) Draw a full ER diagram for the following requirements: The car dealership database stores details considering the following requirements:
 - A salesperson may sell many cars, but each car is sold by only one salesperson.
 - A customer may buy many cars, but each car is bought by only one customer.
 - A salesperson writes a single invoice for each car he or she sells.
 - A customer gets an invoice for each car he or she buys.
 - A customer may come in just to have his or her car serviced; that is, a customer need not buy a car to be classified as a customer.
 - When a customer takes one or more cars in for repair or service, one service ticket is written for each car.
 - The car dealership maintains a service history for each of the cars serviced. The service records are referenced by the car's serial number.

- A car that is serviced may or may not need parts (e.g., adjusting a carburetor or cleaning a fuel injector nozzle does not require providing new parts)
 - Add your own 3-4 meaningful attributes based on common sense
 - Mark relationships clearly (1 to 1, 1 to many, etc.)
3. (25 pts) Using the following tables in the movie database, write generic SQL queries (no need to execute your codes on actual SQL server since you don't have actual data.).
- *Actor* (id, fname, lname, age, gender, nationality)
 - *Movie* (id, name, year, rank, revenue, studio)
 - *Director* (id, fname, lname, gender, income)
- id* column in ACTOR, MOVIE & DIRECTOR tables is a key for the respective table.
- 1) (3 pts) List all the male actors (i.e., gender = 'M').
 - 2) (3 pts) Find the actor whose first name is 'John' and from 'Spain'.
 - 3) (3 pts) List first name and last name of all the actors whose nationality is 'USA'.
 - 4) (3 pts) List the name and revenue of movies made by "Universal Studio".
 - 5) (3 pts) List the total number of movies released in 2018.
 - 6) (3 pts) List max income of all directors.
 - 7) (3 pts) List the name of movies in descending order of revenue, i.e., the highest first.
 - 8) (4 pts) List the average income of female directors.
4. (15 pts) We have the following design of a table to describe student and course information. The primary key is (StudentID, CourseCode). Answer in the following order.

StudentID	StudentName	CourseCode	CourseName	Instructor	InstructorPhone
S001	Alice Smith	CS101	Intro to CS	Dr. Jones	555-1234
S001	Alice Smith	MATH101	Calculus I	Dr. Lee	555-5678
S002	Bob Ray	CS101	Intro to CS	Dr. Jones	555-1234

- 1) (5 pts) Is it in 1NF? Otherwise, make it in 1NF.
- 2) (5 pts) Based on your answer in 1), is it in 2NF? Otherwise, make it in 2NF.
- 3) (5 pts) Based on your answer in 2), is it in 3NF? Otherwise, make it in 3NF.